

DEPARTMENT OF CSE
ASSIGNMENT QUESTIONS

Academic Year : 2026-2027
Batch : 2025-2029
Course Code & Name : 23IT3402& Database Management System
Year/Semester/Section : II/ III /A & B
Regulation : 2023
Subject In charge : Mrs. R .Bharathi, AP/IT
 Ms. R. Anandha Jothi, AP/IT

Total Marks : 80

Q.No	Unit No.	Assignment Task/Questions	Marks	CO	PO
1. Introduce the fundamental concepts of software engineering. 2. Aid in development of skills necessary to elucidate, analyze and document requirements 3. Introduce the basic concepts of various software design approaches and templates 4. Understand the concepts of different testing methods 5. Analyze different metrics for efficient software project management					
1.	UNIT 1	A government healthcare organization is developing a centralized patient-information database to provide services to hospitals in urban and rural areas. The database stores patient records, medical history, medicine availability and treatment information. The system must support multiple users, protect sensitive patient information, minimize redundant data storage and provide controlled access to different categories of users. As a database designer, select appropriate DBMS architecture, relational model, keys, views and SQL access mechanisms for the system. For a given scenario where doctors, pharmacists, administrators and patients require different levels of access, determine an appropriate database design and justify how the design improves data privacy, social accessibility, resource efficiency and collaborative development. Also propose a small project plan showing how the database can be developed and deployed by a team.	16	CO1	PO6, PO7, PO8, PO9, PO10, PO11
2.	UNIT 2	A university maintains a database containing STUDENT, COURSE, FACULTY, DEPARTMENT, ENROLLMENT and PROJECT information. The initial relation contains student details, course details, faculty information and project information in a single large relation, resulting in	16	CO2	PO5, PO6, PO8, PO9, PO10



		insertion, deletion and update anomalies. The following functional dependencies are identified: <i>StudentID</i> → <i>StudentName, Department</i> <i>CourseID</i> → <i>CourseName, Credits, FacultyID</i> <i>FacultyID</i> → <i>FacultyName, Department</i> <i>(StudentID CourseID)</i> → <i>Grade</i> Using the above information, determine a lossless and dependency-preserving decomposition into appropriate normal forms and identify the highest normal form achievable for the resulting relations. Then select an appropriate database-design tool for implementing the ER/EER model and discuss how the design can support fair access to academic information, protection of student data, team-based development and effective technical communication.			
3.	UNIT 3	An online banking system executes the following transactions concurrently: $T_1: R(A), R(B), B = B - 100, W(B), W(A)$ $T_2: R(A), R(B), A = A + 100, W(A), W(B)$ where <i>A</i> and <i>B</i> represent two bank accounts. During execution, the system experiences concurrent updates and a failure occurs immediately after one transaction writes its updated value. Analyse the given schedule to determine whether it is conflict serializable. Select an appropriate locking protocol and isolation level to prevent inconsistent account balances, identify the possibility of deadlock, and determine the recovery mechanism required after failure. Finally, design a team-based transaction-management strategy that considers reliability of financial services, ethical handling of customer data, resource efficiency, communication among development teams and project-risk management.	16	CO3	PO6, PO7, PO8, PO9, PO10, PO11
4.	UNIT 4	An e-commerce company maintains 100 million customer and transaction records. The database must support frequent searches using <i>Customer_ID</i>, range queries on <i>Transaction_Date</i>, and large analytical queries. The organization needs high availability and must continue operation even when disk failures occur. For the following storage alternatives:	16	CO4	PO5, PO6, PO7, PO8, PO9, PO10, PO11



		• RAID 0 • RAID 1 • RAID 5 • RAID 6 and indexing alternatives: • B-tree • B+ tree • Static hashing • Dynamic hashing determine the most appropriate combination of RAID level, file organization and indexing technique for the given workload. Justify the selection using storage efficiency, fault tolerance, search complexity and query-processing requirements. Further, propose a cost-aware query optimization strategy and prepare a team implementation plan considering energy consumption, ethical handling of customer data, modern database tools and project-resource constraints.			
5.	UNIT 5	A social-networking application stores billions of user profiles, posts, friendships and interactions. The system receives millions of transactions per day and requires high scalability, distributed storage and fast retrieval. At the same time, users expect their personal information to be protected. The development team is considering MongoDB, Cassandra and a Graph Database. Select the most appropriate database technology for each of the following operations: 1. Storing flexible user profiles. 2. Handling high-volume distributed writes. 3. Finding multi-hop relationships such as “<i>friends of friends who interacted with a particular product.</i>” 4. Implementing access control for different categories of users. 5. Preventing SQL/NoSQL injection and unauthorized access.	16	CO5	PO5, PO6, PO8, PO9, PO10



		Justify each selection and design an appropriate security and access-control strategy. Finally, formulate a team-based implementation approach that considers privacy, ethical use of user data, scalability, communication among team members and selection of appropriate modern database tools.			
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Dead Line : Submit it on or before 15.09.2026

COURSE IN-CHARGE

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